

Resolution Geometry

The Theorem Ladder

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Abstract

Resolution Geometry (RG) is the geometry of what a declared family of observation protocols can distinguish. Its primitive is the equality-of-law relation of a joint observation law. The relation defines a kernel groupoid and a quotient; the stabilizer of the law modulo its gauge part defines an effective symmetry group acting on the quotient. On regular strata every structure computed from laws descends to the quotient and is invariant under that group. Across walls where distinguishability is lost, the structure continues by non-invertible specialization: kernels are upper semicontinuous, quotients specialize under kernel inclusion, symmetries descend through the normalizer of the wall kernel, and distinguishability specializes under Hellinger convergence. Differential, statistical, causal, temporal, dynamical and metric structure are added in a fixed order, each either forced by the preceding objects, imported from established mathematics, or conditioned on a named hypothesis. This paper states the complete ladder from the exact kernel to calibrated physical distance, with proofs of the RG-specific results and explicit boundaries. The quantum metric module is fixed by the declared task and identified from data. The latent connection is not law-natural, and it is canonical under declared second-order latent data. No internal open problem remains; the remaining problems are external or extensions.

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1 Conventions

Scope. Unless stated otherwise, latent spaces are finite-dimensional smooth manifolds, experiments are classical, dominated and regular in the sense of Section 5, and all statements are local to a stratum. Singular loci are treated by stratification.

Status of results. Each result carries one status.

- *Theorem, Proposition, Lemma*: proved here from the preceding objects.
- *Imported Theorem*: an established result, cited and not reproved.
- *Hypothesis*: a named condition H -. . . ; a result that uses it states it.
- *No-Go*: a stronger statement is false; a counterexample is given.
- *Open Problem*: no result is asserted.

A result is called Γ -*conditional* when its hypothesis is that a stated subgroup belongs to the effective symmetry group of Definition 4.1. Such a hypothesis is tested by equality of laws.

Hypotheses used in this paper.

H-sound	declared groupoid lies in the kernel groupoid (Def. 4.5)
H-compl	declared groupoid generates every fibre (Def. 4.5)
H-coh	declared groupoid is normalized by the declared physical group (Prop. 4.8)
H-reg	regular stratum: submersive quotient, immersive descended law (Def. 5.1)
H-nuis	constant target and nuisance rank, or a declared nuisance bound (Hyp. 6.8)
H-sens	compatible sensor family (Def. 7.1)
H-top	topology and regularity for a cone boundary (Hyp. 9.2)
H-int	exact scale cocycle (Hyp. 10.1)
H-SI	SI metrological anchor, with a non-scale-covariant count-loss model (Hyp. 12.10)
H-med	propagation speed anchored or tomographically resolved (Hyp. 12.13)
H-val	held-out anchor validation (Hyp. 12.17)
H-unif	declared locally uniform convergence of a family of laws, with a modulus (Hyp. 5.7)
H-hell	locally uniform Hellinger convergence of a family (Hyp. 6.12)
H-stat	stationarity of a source law across an anchor outage (Hyp. 12.15)
H-task	declared operational task of a quantum protocol (Hyp. 6.16)
H-cov	declared latent covariance and, for dynamics, memory kernel (Hyp. 10.8)
H- Γ	a stated subgroup belongs to Γ (Def. 4.1)

2 Objects

Definition 2.1 (Observation system). An *observation system* is a triple $\mathcal{S} = (X, \{F_i\}_{i \in I}, \Psi)$ consisting of

1. a latent space X ;
2. protocols $F_i : X \rightarrow Y_i$, where Y_i is a space of observable laws (probability laws of records, or deterministic records);
3. a declared relabeling group $\Psi = \prod_i \Psi_i$, where Ψ_i acts on Y_i by pushforward along admissible bijections of outcomes.

The *joint law* is $F = (F_i)_{i \in I} : X \rightarrow Y = \prod_i Y_i$. A single experiment is the case $|I| = 1$.

Definition 2.2 (Counted channel). A protocol is *counted* if its records take values in a finite chain $\{0, 1, \dots, N\}$ or in $\mathbb{N}$, and its admissible relabelings preserve order.

Definition 2.3 (Natural module). A *natural module* is a pullback $M = F^* \mu$ of a Ψ -invariant structure μ on Y : a function, a tensor field, a cone field, a family of functions with transition maps, or a transport defined from laws. A structure on X that is not of this form is *latent*.

Example 2.4. Observable invariants, clock readings, record counts and the Fisher tensor are natural. A transport generated by a latent dynamical law is latent unless it is defined from laws.

3 The exact layer

Theorem 3.1 (Kernel pair and quotient). *The relation $x \sim x' \iff F(x) = F(x')$ is an equivalence relation. Its graph*

$$\mathcal{K} = X \times_Y X \rightrightarrows X, \quad s(x, x') = x, \quad t(x, x') = x', \quad (x', x'') \circ (x, x') = (x, x''),$$

is a groupoid, the kernel groupoid. The quotient $\pi : X \rightarrow Q = X/\mathcal{K}$ satisfies $Q \cong \text{im } F$ canonically.

Proof. Reflexivity, symmetry and transitivity are inherited from equality in Y , and these are the groupoid axioms for the graph. The map $[x] \mapsto F(x)$ is well defined and bijective onto $\text{im } F$. $\square$

Theorem 3.2 (Universal factorization). *A map $H : X \rightarrow Z$ is constant on $\mathcal{K}$ -orbits iff there is a unique $\bar{H} : Q \rightarrow Z$ with $H = \bar{H} \circ \pi$. In particular the observable invariants are exactly the functions on Q .*

Proof. Set $\bar{H}([x]) = H(x)$. It is well defined by fibre constancy and unique because π is surjective. $\square$

Proposition 3.3 (Joint kernel). $\mathcal{K} = \bigcap_i \mathcal{K}_{F_i}$. *The induced map $Q \rightarrow \prod_i X/\mathcal{K}_{F_i}$ is injective and in general not surjective.*

Proof. Equality in a product is equality in every factor. For non-surjectivity take $X = \{1, 2\}$, $F_1 = F_2 = \text{id}$: the product of quotients has four points, Q has two. $\square$

Proposition 3.4 (Garbling). *Let M be a Markov kernel independent of x . Then $\mathcal{K}_F \subseteq \mathcal{K}_{M \circ F}$, with strict inclusion in general.*

Proof. $P_x = P_{x'}$ implies $MP_x = MP_{x'}$. Merging two outcomes identifies $(0.1, 0.3, 0.6)$ with $(0.3, 0.1, 0.6)$. $\square$

No-Go 3.5 (Limits of the exact layer). The kernel groupoid and the quotient do not determine a topology, a differential structure, a metric, a statistical order, a reachability order, a cone, a scale, a connection, a clock, or a unit. Each of these is added in a later section.

4 The symmetry layer

Definition 4.1 (Stabilizer, gauge, effective symmetry).

$$\text{Stab} = \{(\varphi, \psi) \in \text{Diff}(X) \times \Psi : F \circ \varphi = \psi \circ F\}, \quad \text{Gau} = \{\varphi \in \text{Diff}(X) : F \circ \varphi = F\}.$$

Pairs whose relabelings agree on $\text{im } F$ are identified. The *effective symmetry group* is $\Gamma = \text{Stab}/\text{Gau}$.

Theorem 4.2 (Kernel exact sequence). *1. Every element of Stab maps $\mathcal{K}$ to itself: $(\varphi \times \varphi)(\mathcal{K}) = \mathcal{K}$.*

2. Gau is a normal subgroup of Stab.

3. Γ acts faithfully on Q by $[\varphi] \cdot [x] = [\varphi x]$, and

$$1 \longrightarrow \text{Gau} \longrightarrow \text{Stab} \longrightarrow \Gamma \longrightarrow 1$$

is exact.

Proof. If $F(x) = F(x')$ then $F(\varphi x) = \psi F(x) = \psi F(x') = F(\varphi x')$; this gives (1) and a well-defined action on Q . An element acts trivially on Q iff $F \circ \varphi = F$, i.e. iff it lies in Gau; this gives normality and faithfulness. $\square$

No-Go 4.3 (Non-splitting). The sequence of Theorem 4.2 need not split. For $X = \mathbb{Z}_4$, $F(x) = x \bmod 2$ and the declared group $\mathbb{Z}_4$ of rotations, the gauge part is $\{0, 2\}$ and the image in $\text{Aut}(Q)$ is $\mathbb{Z}_2$; the only subgroup of order two is the gauge part itself. An observable symmetry therefore need not lift to a latent symmetry of the same order.

Lemma 4.4 (Count rigidity). *The order-preserving bijections of a finite chain are the identity. Hence Ψ_i is trivial on every counted channel, and no element of Stab rescales a counted record.*

Proof. An order-preserving bijection of $\{0, \dots, N\}$ fixes 0 and, inductively, every successor. $\square$

Definition 4.5 (Declared assembly). A *declared groupoid* is a groupoid $A \rightrightarrows X$. It is *sound* (**H-sound**) if $A \subseteq \mathcal{K}$ and *complete* (**H-compl**) if $A \cdot x = [x]$ for all x . A *declared physical group* is a subgroup $G \leq \text{Stab}$ with image $\Gamma_G \leq \Gamma$.

Theorem 4.6 (Orbit–quotient comparison). *Under H-sound the map $X/A \rightarrow Q$, $[x]_A \mapsto [x]$, is a well-defined surjection. It is a bijection iff H-compl holds.*

Proof. Soundness gives well-definedness; surjectivity of π gives surjectivity; injectivity is the statement that each fibre is one A -orbit. $\square$

No-Go 4.7 (Saturation hierarchy). The following are distinct: (a) full differential rank of a family of A -invariants; (b) global separation of A -orbits by those invariants; (c) completeness; (d) reconstruction of a latent representative. The map $(u, v) \mapsto (u^2, v)$ has full generic rank and identifies $(\pm u, v)$; a coordinate absent from F and not moved by A produces several A -orbits in one fibre; an orbit does not select a representative.

Proposition 4.8 (Coherence). *If $gAg^{-1} = A$ for all $g \in G$ (H-coh), the completeness locus $\{x : A \cdot x = [x]\}$ is G -invariant. If Γ_G acts transitively on a stratum, completeness at one point of the stratum implies completeness on the stratum. Without H-coh completeness can depend on the point: on $\mathbb{Z}_6 \times \mathbb{Z}_2$ with $F(x, b) = x$, bit flips declared at even x are sound and complete exactly at even x .*

Proof. $g(A \cdot x) = (gAg^{-1})(gx) = A \cdot gx$, and $g[x] = [gx]$ by Theorem 4.2. $\square$

5 The differential layer

Definition 5.1 (Regular stratum). A *regular stratum* is an open $\mathcal{K}$ -saturated submanifold $S \subseteq X$ such that $Q_S = S/\mathcal{K}$ is a manifold, $\pi|_S$ is a submersion, and the descended law $\bar{F} : Q_S \rightarrow Y$ is an immersion into a smooth model of law space (**H-reg**). The *observational vertical space* is $V_x = \ker D\pi_x$.

Theorem 5.2 (Tangent sequence). *On a regular stratum*

$$0 \longrightarrow V \longrightarrow TS \xrightarrow{D\pi} \pi^*TQ_S \longrightarrow 0$$

is exact, so $T_{[x]}Q_S \cong T_xX/V_x$. For a sound declared groupoid, $V_x^A = T_x(A \cdot x) \subseteq V_x$, and V_x/V_x^A is the infinitesimal completeness defect.

Theorem 5.3 (Natural descent). *On a regular stratum let $M = F^*\mu$ be a natural module. Then*

1. $M = \pi^*\bar{M}$ with $\bar{M} = \bar{F}^*\mu$ unique; M annihilates V and is independent of the representative in each fibre;
2. $\bar{M}$ is Γ -invariant;
3. if M is a symmetric tensor and μ is nondegenerate, $\text{rad } M = V$.

Proof. (1) $F^*\mu = (\bar{F} \circ \pi)^*\mu = \pi^*\bar{F}^*\mu$. (2) For $(\varphi, \psi) \in \text{Stab}$, $\varphi^*F^*\mu = (\psi \circ F)^*\mu = F^*\psi^*\mu = F^*\mu$. (3) $\bar{F}$ is immersive and μ nondegenerate, so $\bar{M}$ is nondegenerate and the radical of $\pi^*\bar{M}$ is $\ker D\pi$. $\square$

Corollary 5.4 (Automatic descent). *For natural modules, basicness with respect to $\mathcal{K}$, representative independence, and projectability of law-defined transports and law-defined cone fields hold without further hypothesis. For latent structures they are hypotheses.*

Imported Theorem 5.5 (Singular strata; [12, 15, 49, 7]). *Orbit spaces of proper Lie groupoids carry orbit-type stratifications with slice models. At rank, isotropy or orbit-type changes the single tangent quotient is replaced by slice data and tangent cones.*

Proposition 5.6 (Tangent cone at an isotropy wall). *For $\mathbb{Z}_2$ acting by $(u, v) \mapsto (u, -v)$ and $\pi(u, v) = (u, v^2)$, $D\pi_{(0,0)}$ has rank one, while jet-rescaled secants $(\Delta u/h, \Delta r/h^2)$ fill $\mathbb{R} \times \mathbb{R}_{\geq 0}$. The first-order tangent misses a one-sided observable direction; higher jets of F are required at walls.*

5.1 Specialization across walls

A wall is reached either by a family of joint laws $F_\varepsilon \rightarrow F_0$ (a protocol wall) or along $x_n \rightarrow x$ inside one law (a latent wall). Write $\mathcal{K}_\varepsilon, Q_\varepsilon, V_\varepsilon$ for the objects of F_ε .

Hypothesis 5.7 (H-unif). $F_\varepsilon \rightarrow F_0$ locally uniformly, with a declared modulus (for instance a uniform Lipschitz bound) that is checked at every resolved scale.

Theorem 5.8 (Kernel upper semicontinuity). *Under H-unif, with F_0 continuous, $\limsup_{\varepsilon \rightarrow 0} \mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$ (Kuratowski upper limit).*

Proof. If $(x_\varepsilon, x'_\varepsilon) \in \mathcal{K}_\varepsilon$ converge to (x, x') , then $F_0(x) - F_0(x')$ is a limit of $F_\varepsilon(x_\varepsilon) - F_\varepsilon(x'_\varepsilon) = 0$, by uniform convergence and continuity of F_0 . $\square$

No-Go 5.9 (Pointwise convergence and sampling; [19]). $F_\varepsilon = \text{id} + B_\varepsilon$, with a bump of height 1 and width ε^2 at $x = \varepsilon$, converges pointwise, but $(\varepsilon, 1 + \varepsilon) \in \mathcal{K}_\varepsilon$ tends to $(0, 1) \notin \mathcal{K}_0$. A sampling grid reports sup-norm 0 for this family. Locally uniform convergence is therefore declared (H-unif), not certified by sampling.

Proposition 5.10 (Specialization map). *1. A map $\sigma : Q_\varepsilon \rightarrow Q_0$ with $\pi_0 = \sigma \circ \pi_\varepsilon$ exists iff $\mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$; it is then unique and surjective.*

2. Every garbling wall has such a map (Proposition 3.4). Without inclusion, specialization is the correspondence $\limsup \mathcal{K}_\varepsilon$; for example, $x^2 + \varepsilon x$ identifies $(x, -x - \varepsilon)$, which x^2 separates.

3. Specializations compose: $\sigma_{\delta 0} \circ \sigma_{\varepsilon \delta} = \sigma_{\varepsilon 0}$.

4. Exactly the $\mathcal{K}_0$ -basic invariants descend.

Proof. (1) and (4) are Theorem 3.2. (3) follows from uniqueness of the factorization. $\square$

Theorem 5.11 (Symmetry descent). *An element of $\text{Stab}(F_\varepsilon)$ induces a map of Q_0 iff it maps $\mathcal{K}_0$ to itself. The descended image is a subgroup of $\text{Aut}(Q_0)$ and can be a proper subgroup of Γ_0 .*

Proof. $[x]_0 \mapsto [\varphi x]_0$ is well defined iff φ preserves $\mathcal{K}_0$ -classes. Strictness: generic regular laws can specialize to cyclically related wall laws, so the descended image is trivial while $\Gamma_0 \supseteq \mathbb{Z}_3$ [19]. $\square$

Theorem 5.12 (Tangent upper semicontinuity). *If $F_\varepsilon \rightarrow F_0$ in C^1 (or $x_n \rightarrow x$ within one C^1 law), every Grassmann limit point of $V_\varepsilon(x_\varepsilon)$ lies in $V_0(x)$, and $\text{rank } DF_0(x) \leq \liminf \text{rank } DF_\varepsilon(x_\varepsilon)$. For tame laws the rank strata are finite [33].*

Proof. For unit $v_\varepsilon \in V_\varepsilon$ with $v_\varepsilon \rightarrow v$, $DF_0 v = \lim DF_\varepsilon v_\varepsilon = 0$. Nonvanishing minors persist. $\square$

Definition 5.13 (Stratified quotient category). The objects are the quotients of the strata. The arrows are the invertible transitions inside a stratum, together with the specializations of Proposition 5.10 (maps, or correspondences). A path is admissible if every edge is.

Theorem 5.14 (Stratified structure). *Under H-unif the stratified quotient category is a category. A specialization is invertible iff $\mathcal{K}_\varepsilon = \mathcal{K}_0$. Holonomy is defined inside strata; two admissible paths into a wall are compared after specialization, $\sigma \circ P_1$ against $\sigma \circ P_2$, and differences inside the gained kernel are invisible.*

Proof. Composition of maps and of closed relations is associative. Injectivity of σ holds iff no new identifications occur. Comparison in Q_0 quotients by $\mathcal{K}_0$. $\square$

6 The second-order layer

Definition 6.1 (Fisher module). On a dominated regular experiment with score $S_u = \partial_u \log p(y | x)$, the Fisher tensor is $g_F(u, w) = \mathbb{E}[S_u S_w]$. It is the pullback of the Fisher–Rao metric along the law map and is a natural module.

Proposition 6.2 (Fisher descent and ignorance). *On a regular stratum $g_F = \pi^* \bar{g}_F$, $\bar{g}_F$ is Γ -invariant, and $\text{rad } g_F = V$. For $Y = Jx + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \Sigma)$, $\Sigma \succ 0$: $g_F = J^\top \Sigma^{-1} J$ and $\text{rad } g_F = \ker J$.*

Proof. Theorem 5.3 with μ the Fisher–Rao metric; the Gaussian formula is standard. $\square$

Imported Theorem 6.3 (Regular reconstruction; [45, 48]). *The resolved sector $R = \text{im } J$, the null sector $N = \ker J^*$ and the covariance block geometry are gauge-equivariant functions of (J, Σ, g_O) ; the Fisher tensor is the inverse Schur complement of the covariance on the resolved block.*

No-Go 6.4 (Fisher insufficiency). g_F does not determine the covariance. Covariances with equal Schur complement on the resolved block and different null blocks have the same Fisher tensor.

Proposition 6.5 (Second-order quotient; [44]). *Let $O(p) \times O(q)$ act on triples (Σ_R, Σ_N, C) . On the regular stratum the orbit space has dimension $p + q + pq$. The two spectra and the $\min(p, q)$ canonical correlations therefore do not coordinate it when $pq > \min(p, q)$; the mixed traces $\text{tr}(\Sigma_R^a C \Sigma_N^b C^\top)$ over finite exponent ranges supply generically independent coordinates, up to a finite ambiguity.*

Proof of the dimension. Ambient dimension $\frac{p(p+1)}{2} + \frac{q(q+1)}{2} + pq$ minus group dimension $\frac{p(p-1)}{2} + \frac{q(q-1)}{2}$, with finite generic stabilizer. $\square$

No-Go 6.6 (Shadow is not law). Distinct laws can share means and covariances (higher-moment escape), and processes with equal one-time covariance can differ in their two-time laws (memory escape). The second-order shadow is a natural module of the law, not the law.

Definition 6.7 (Efficient metric). For target coordinates q and nuisance coordinates ν , $g_{\text{eff}} = I_{qq} - I_{q\nu} I_{\nu\nu}^+ I_{\nu q}$.

Hypothesis 6.8 (H-nuis). Target and nuisance ranks are constant on the stratum, or a bound on the nuisance is declared and included in the efficient metric.

Proposition 6.9 (Nuisance conventions). g_{eff} is invariant under $\nu \mapsto \nu + f(q)$ and depends on the declared target chart. Without H-nuis it can be discontinuous: for $I_\varepsilon = \begin{pmatrix} a & \rho\sqrt{a\varepsilon} \\ \rho\sqrt{a\varepsilon} & \varepsilon \end{pmatrix}$, $\lim_{\varepsilon \rightarrow 0^+} g_{\text{eff}} = a(1 - \rho^2)$ whereas $g_{\text{eff}} = a$ at $\varepsilon = 0$.

Imported Theorem 6.10 (Data processing; [5, 1]). *For a parameter-independent Markov kernel, $g_{F, M \circ F} \preceq g_{F, F}$.*

No-Go 6.11 (Kernel order is not Blackwell order; [3, 11]). Two injective experiments have the same kernel (the diagonal) but need not be Blackwell equivalent: $\mathcal{N}(x, 1)$ and $\mathcal{N}(x, 4)$ are both injective and the second is a strict garbling of the first.

Hypothesis 6.12 (H-hell). A family $F_\varepsilon \rightarrow F_0$ is locally uniformly Hellinger-continuous; for dominated regular families with uniformly differentiable square-root densities this is Le Cam convergence of the local experiments [34].

Theorem 6.13 (Specialization of distinguishability). *Under H-unif and H-hell, Hellinger distances, natural Fisher modules and Bayes risks converge to their wall values, $\text{rad } g_0 \supseteq \limsup V_\varepsilon$, and the Cramér–Rao bound diverges on the gained kernel.*

Proof. The Hellinger triangle inequality gives distances; the second-order Hellinger expansion gives Fisher; total variation is bounded by Hellinger distance, which gives risks. $\square$

No-Go 6.14 (Weak convergence is not enough). For y observed as $\mathcal{N}(\varepsilon y, \varepsilon^2)$, the laws converge weakly to a wall law identifying y . Yet the Fisher information in y stays 1, the Hellinger distance between $y = 0.3$ and $y' = 1.1$ stays 0.277, and the minimal error of testing $y = 0$ against $y = 1$ stays 0.309 instead of $1/2$.

6.1 The quantum metric module

For finite-dimensional quantum protocols, $F(\theta)$ is the joint outcome law of a declared measurement family on ρ_θ . When the family is tomographically complete, the kernel is equality of states. Natural metric modules must be monotone under CPTP maps.

Imported Theorem 6.15 (Petz classification; [22]). *The CPTP-monotone Riemannian metrics on full-rank states are*

$$c g_f(\rho; A, A) = c \sum_{i,j} \frac{|A_{ij}|^2}{\lambda_j f(\lambda_i/\lambda_j)},$$

with f operator monotone, $f(t) = tf(1/t)$, $f(1) = 1$ and $c > 0$. On commuting tasks all g_f equal the classical Fisher metric. Monotonicity is therefore a family gate, not a selector.

Hypothesis 6.16 (H-task). The quantum protocol declares its operational task (estimation, asymmetric testing, symmetric testing, or another declared local task).

Theorem 6.17 (Task lock; [23, 24, 25, 26, 27, 20]). *Under H-task, the task fixes the Petz member up to scale:*

- *optimal-measurement estimation gives SLD, since $\max_{\text{POVM}} F_{\text{cl}} = g_{\text{SLD}}$;*
- *Stein testing gives BKM, the Hessian of the Umegaki relative entropy;*
- *Chernoff testing gives Wigner–Yanase, since $\xi = h^2 g_{\text{WY}}/8$ locally.*

Theorem 6.18 (Nonparametric identification; [20]). *Commuting tasks fix the scale. For $\rho = \text{diag}(\lambda_1, \lambda_2)$ and $A = \sigma_x/\sqrt{2}$, a response y determines $f(\lambda_1/\lambda_2) = 2\kappa|A_{12}|^2/(\lambda_2 y)$. Dense ratios determine f . Finite-panel fits are not locks; a metric certificate is a band of f of width set by noise divided by the response gap.*

Theorem 6.19 (Quantum wall specialization; [20]). *On tangents coupling support and kernel at a support wall, g_f specializes iff $f(0) > 0$. SLD tends to $4 \times$ Fubini–Study. BKM, Heinz(1/4) and RLD diverge like $\log(1/\varepsilon)$, $\varepsilon^{-1/4}$ and ε^{-1} .*

Wall geometry is obtained by re-discrimination of the specialized task, with outcomes LOCKED, DIVERGENT, VACUOUS, UNDERDETERMINED and INCONSISTENT. Metric selection selects no transport.

7 Ignorance and the observation limit

Definition 7.1 (Persistent null). For a family $\{F_\rho\}$ of protocols on a common latent space (H-sens), the persistent null at x is $\mathcal{I}_x = \bigcap_\rho \ker DF_\rho|_x$.

Proposition 7.2 (Finite attainment). *In finite dimension $\mathcal{I}_x$ equals the intersection over finitely many members of the family, and it equals V_x of the joint law of that finite subfamily on its regular stratum.*

Proof. A descending chain of subspaces of a finite-dimensional space stabilizes. The joint state-ment is Proposition 3.3 at the tangent level. $\square$

Proposition 7.3 (Three layers). *For smooth maps E (evaluation at finite precision) and R (downstream representation), the exact null of the sensor is $\ker DF$. The precision loss of E and the representation loss of R are nulls of the composites $E \circ F$ and $R \circ E \circ F$ and satisfy $\ker DF \subseteq \ker D(E \circ F) \subseteq \ker D(R \circ E \circ F)$. Only $\ker DF$ is a property of the sensor.*

Imported Theorem 7.4 (Finite-sample boundary reconstruction; [43]). *For a correctly specified regular protocol with a design that places positive mass on both sides of an operational boundary and a nonsingular Fisher matrix, the maximum-likelihood boundary estimator is consistent and asymptotically normal with covariance given by the inverse Fisher matrix. The quotient alone determines no numerical boundary: affine recalibration of the latent coordinate moves it.*

Proposition 7.5 (Rank walls are not certified from finite data). *A rank certificate is FULL when a lower confidence bound of $\sigma_{\min}$ exceeds a declared τ at sample sizes n and $4n$; otherwise only an upper bound on $\sigma_{\min}$ is reported. A true wall and a nearby full-rank law with $\sigma_{\min} < \tau$ give the same certificate [19].*

8 Statistical distance and angle

Definition 8.1 (RG length, distance, angle). On a regular stratum on which $\bar{g}_{\text{eff}}$ is positive definite,

$$L[\gamma] = \int \sqrt{\bar{g}_{\text{eff}}(\dot{\gamma}, \dot{\gamma})} dt, \quad d_{RG}(q_0, q_1) = \inf_{\gamma} L[\gamma], \quad \cos \angle(u, w) = \frac{\bar{g}_{\text{eff}}(u, w)}{\|u\| \|w\|}.$$

Theorem 8.2 (Distance law). *Under H -reg and H -nuis, d_{RG} is the Riemannian distance of $\bar{g}_{\text{eff}}$ on Q_S . It is independent of the latent representative and of the latent chart, Γ -invariant, and dimensionless.*

Proof. Theorem 5.3 and Proposition 6.2; the Fisher tensor is invariant under reparametrization and the score is dimensionless. $\square$

Proposition 8.3 (Monotonicity). *Under a parameter-independent garbling, path lengths and d_{RG} do not increase.*

No-Go 8.4 (Rank one). On a rank-one stratum, length and order along the resolved direction are defined, and no angle between independent directions is.

No-Go 8.5 (Finite divergences). Only the intrinsic length is additive under concatenation. For the Gaussian location family the Bhattacharyya angle between p and $p(\cdot - a)$ is $2 \arccos e^{-a^2/8\sigma^2}$, which is not linear in a and is bounded by π .

No-Go 8.6 (Experiment relativity). For N samples of $\mathcal{N}(x, \sigma^2)$, $d_{RG} = \sqrt{N} |\Delta x| / \sigma$. On a class of experiments closed under additive-noise garbling there is no function f with $f(d_{RG}) = |\Delta x|$: $d_{RG}(\Delta x = 1, \tau = 0) = d_{RG}(\Delta x = 2, \tau = \sqrt{3}\sigma)$ for added noise of variance τ^2 .

9 Order, cone and conformal class

Definition 9.1 (Reachability). A declared preorder $\preceq$ on X is *fibre-compatible* if it is constant on $\mathcal{K}$ -orbits. It descends to Q ; its antisymmetrization is the causal order.

Hypothesis 9.2 (H-top). Q_S carries a topology compatible with $\preceq$, and the future sets have a regular boundary whose tangent directions form a cone field $C_q \subset T_q Q_S$.

No-Go 9.3 (Projective typing). The realization type of C_q is determined by its projective boundary, not by a defining function: $\{Q = 0\}$ and $\{Q^3 = 0\}$ define the same cone.

Theorem 9.4 (Realization branches). *Under H -top the first applicable case describes C_q on a stratum:*

1. *the null cone of a nondegenerate Lorentzian quadratic form: then C determines a positive conformal ray $[g]$ (Imported Theorem 9.5);*
2. *a single smooth, strongly convex, homogeneous cone with Lorentzian fibre Hessian: a Lorentz–Finsler structure;*
3. *several persistent characteristic sheets: a multicone or multi-Finsler structure;*
4. *a cone whose rank, signature, topology, isotropy or sheet count changes: stratified realization by slices and tangent cones (Imported Theorem 5.5);*
5. *otherwise: general characteristic or admissibility geometry.*

Imported Theorem 9.5 (Conformal reconstruction; [8, 13]). *Two nondegenerate Lorentzian quadratic forms on a real vector space with the same null cone are proportional; a time orientation fixes the sign. Globally, under distinguishing causality, the causal structure of a Lorentzian manifold determines its conformal class and not its conformal factor.*

Theorem 9.6 (Lorentz isotropy forces the quadratic branch). *Suppose the isotropy Γ_q acts on $T_q Q \cong \mathbb{R}^{1+n}$ and contains $SO^+(1, n)$ in its standard representation, $n \geq 2$ (H - Γ). Then every Γ_q -invariant closed convex pointed cone with nonempty interior is the future or the past light cone, and the Γ_q -invariant symmetric bilinear forms are $\{\lambda\eta : \lambda \in \mathbb{R}\}$. For $n = 1$ the conclusion fails: the wedge $\{x \geq |t|\}$ is invariant.*

Proof. An invariant cone is a union of orbits: sheets of two-sheeted hyperboloids, null half-cones, one-sheeted hyperboloids and the origin. For $n \geq 2$ a one-sheeted hyperboloid contains v and $-v$ and is excluded by pointedness. Invariance under positive scaling, convexity and closedness leave one closed light cone. The standard representation is absolutely irreducible, so invariant symmetric forms form a line. $\square$

9.1 The residual characteristic branch

Theorem 9.7 (Tame reduction; [40, 33, 18]). *If $\mathbb{P}A$ is definable in an o-minimal structure, then:*

- *it has a finite canonical Whitney stratification with constant signature of the second fundamental form on each stratum;*
- *it has finitely many sheets ($\leq (d-1)(d-2)/2 + 1$ for a degree- d plane curve, Harnack) and integer dimension;*
- *its finite-resolution component counts stabilize;*
- *its type, consisting of the strata, their incidences and signatures, and the sheet monodromy, is finite combinatorial data. Bounded-complexity families have finitely many types up to stratified homeomorphism.*

No-Go 9.8 (Linear moduli). Polyhedral cones with five extreme rays carry continuous projective invariants. Tame types are classified up to stratified homeomorphism, not up to $GL(V)$.

Imported Theorem 9.9 (Duality; [38]). *A closed convex pointed cone is smooth iff its dual is strictly convex. A Lorentz–Finsler realization requires a primal and a dual that are both smooth.*

Theorem 9.10 (Hyperbolic lock; [31, 32, 37, 18]). *If the characteristic set is the zero set of a principal symbol hyperbolic in a time direction (for constant coefficients, a well-posed Cauchy problem), then:*

- the forward cone is convex;
- the sheets are real, ordered and nested, with trivial label monodromy;
- the only residual structure is contact between sheets, with $\mathbb{Z}_2$ polarization holonomy at conical points.

Conversely, a nonconvex or label-exchanging characteristic set of a principal symbol forces complex roots, with growth rate proportional to frequency (Hadamard ill-posedness).

No-Go 9.11 (Wild class). Non-definable direction sets (fractal, infinitely accumulating) are characterized only by resolution invariants: count exponent, box dimension, Cantor–Bendixson rank. They admit no finite classification, and finite data certify only window statements: no finite window certifies tameness or wildness [18].

10 Scale and transport

Hypothesis 10.1 (H-int). Local scale representatives $g_i = \Omega_i^2 g$ on a cover differ on overlaps by positive functions $\rho_{ij} = \Omega_i/\Omega_j$ forming an exact cocycle, $\rho_{ij} = \sigma_i/\sigma_j$.

Proposition 10.2 (Scale representative). *A global metric representative of $[g]$ exists on a connected stratum iff H-int holds. A nonzero loop product of ρ is scale holonomy: a unit transported around the loop returns multiplied by it. Data along one worldline fix at most one normalization along that worldline.*

Proposition 10.3 (Strict transport). *If two representatives g and $\Omega^2 g$ are both preserved by a strict transport on a transport-connected stratum, Ω is constant there.*

Proof. $P_{qr}^*(\Omega^2 g)_r = (\Omega^2 g)_q$ and $P_{qr}^* g_r = g_q$ give $\Omega(r) = \Omega(q)$ by nondegeneracy. $\square$

Imported Theorem 10.4 (Induced transport). *On a regular embedded metric stratum with orthogonal projector P , the induced connection is $\nabla^W = P\nabla^{LC}$ and the second fundamental form is $(I - P)\nabla^{LC}$.*

No-Go 10.5 (Entropy does not select a connection). $H(X)$, $H(Q)$ and $H(X | Q)$ are invariant under fibre-preserving automorphisms, and horizontal lifts are not; no function of these entropies selects a lift.

Proposition 10.6 (Law-defined transport). *A transport defined from laws descends uniquely to Q_S (Corollary 5.4). A latent transport W descends iff $D\pi(W_x)$ is constant on fibres.*

10.1 Canonical transport

Theorem 10.7 (No law-natural connection; [21]). *For lifts B_1, B_2 with $DF B_i = I$, the vertical shear $\varphi = \text{id} + (B_2 - B_1)DF$ preserves the law and maps B_1 to B_2 . Hence Gau acts transitively on horizontal complements, and no Gau-invariant connection exists when $V \neq 0$. Laws, natural modules, entropy, wall specialization and quantum metric selection therefore select no connection.*

Proof. $DF(B_2 - B_1) = 0$, and $\varphi B_1 = B_1 + (B_2 - B_1)DF B_1 = B_2$. Natural modules are Gau-invariant (Theorem 5.3). $\square$

Hypothesis 10.8 (H-cov). A latent covariance $P \succ 0$ is declared, and for dynamics also a memory kernel (a latent process law).

Theorem 10.9 (Canonical connection; [21, 28, 29, 30]). *Under H-cov, $H_P = \text{im}(P DF^\top) = V^{\perp_{P^{-1}}}$ is a complement to V , and:*

1. it is the unique minimal- P^{-1} -norm lift and the noiseless limit of the Kalman update;
2. it is equivariant, and different declared P give different transports on identical laws;
3. its curvature (the O'Neill tensor) is the latent holonomy of closed observed loops, of order their area, and zero for linear laws;
4. with a memory kernel, the canonical lift is the path-space conditional mean, which equals the RTS smoother;
5. across a rank wall with the same P , $H_{P,0} \subseteq \lim H_{P,\varepsilon}$, and wall transport is the P^{-1} -orthogonal projection onto $H_{P,0}$.

Proof. (1) Minimal-norm solution of $DFx = \delta y$; let $R \rightarrow 0$ in the Kalman gain. (2) Substitution. (3) O'Neill's integrability tensor. (4) Gaussian conditioning on path space. (5) Tangent upper semicontinuity (Theorem 5.12) and reversal of inclusions under complements. $\square$

11 Duration

Let $T_i : H \rightarrow \mathbb{R}$ be the readouts of clock protocols on a history space H , i.e. natural modules of the clock channels.

Theorem 11.1 (Clock transition groupoid). *If $\mathcal{K}_{T_i} = \mathcal{K}_{T_j}$, there is a unique bijection $f_{ji} : T_i(H) \rightarrow T_j(H)$ with $T_j = f_{ji} \circ T_i$, and*

$$f_{ii} = \text{id}, \quad f_{ij} = f_{ji}^{-1}, \quad f_{ki} = f_{kj} \circ f_{ji}.$$

If $\mathcal{K}_{T_i} \neq \mathcal{K}_{T_j}$ no such map exists.

Proof. Theorem 3.2 applied to the common quotient; uniqueness closes composition. If the kernels differ, some pair is identified by one readout and separated by the other. $\square$

Theorem 11.2 (Affine clocks). *Suppose a flow Φ_s belongs to Stab , $T_i \circ \Phi_s = T_i + a_i s$ with $a_i > 0$, and each $T_i(H)$ is an interval $(H\text{-}\Gamma)$. Then every f_{ji} is affine with slope a_j/a_i . The clocks are charts of one duration object, unique up to a positive unit and an origin on each connected component.*

Proof. $f_{ji}(T_i \circ \Phi_s) = T_j \circ \Phi_s$ gives $f_{ji}(t + a_i s) = f_{ji}(t) + a_j s$ for all admissible t, s . $\square$

Proposition 11.3 (Path independence). *An accumulated duration T is a function of a declared increment Δt iff $\mathcal{K}_{\Delta t} \subseteq \mathcal{K}_T$ on H . Unsigned elapsed duration is invariant under path reversal and fixes no time orientation.*

Imported Theorem 11.4 (Coherence duration; [16]). *For multiplicative coherence operators C , $\mathcal{T} = -\log C$ is additive on commuting families, and the defect of additivity is the Baker–Campbell–Hausdorff series $\frac{1}{2}[\mathcal{T}_1, \mathcal{T}_2] + \dots$.*

11.1 Dynamics

For a map f with a finite observation s , the history protocols are $F_n = (s, s \circ f, \dots, s \circ f^{n-1})$. Time translation acts on records by the shift.

Proposition 11.5 (Finite-history regularity; [17]). *For the logistic family with $x_0 \sim U(0, 1)$, the n -word law is continuous in r and smooth off finitely many kneading walls, with finite Fisher–Rao length across them. The noiseless wall (non-domination, discontinuous law map) occurs only as $n \rightarrow \infty$.*

Imported Theorem 11.6 (Kernel-count entropy; [36, 17]). *For a unimodal f and its critical partition, $|X/\mathcal{K}_{F_n}|$ is the lap number of f^n , and $h_{\text{top}} = \lim \frac{1}{n} \log |X/\mathcal{K}_{F_n}| = \inf_n \frac{1}{n} \log |X/\mathcal{K}_{F_n}|$.*

Imported Theorem 11.7 (Entropy locks; [39, 35]). *For ergodic μ of a C^2 interval map, $h_\mu \leq \max(\lambda, 0) \leq h_{\text{top}}$, with $h_\mu = \lambda$ iff μ is absolutely continuous.*

Definition 11.8 (Typed certificate). A dynamical certificate names its module (kernel count, history law, derivative, exact return, resolution) and its escalation window. It is issued only on an escalation plateau and only if it is consistent with Imported Theorem 11.7 and with exact return for periodicity. Zero entropy and a rank wall are never certified from finite data.

Imported Theorem 11.9 (Cone-field criterion; [42]). *A continuous strictly invariant cone field with uniform expansion exists iff the dynamics are uniformly hyperbolic. Nonuniformly hyperbolic characteristic processes are typed per window.*

12 Physical distance

12.1 No internal unit

No-Go 12.1 (Unit). 1. If a readout scale s is not anchored, the law depends on $(x/s, \sigma/s)$, and the dilation $(x, \sigma, s) \mapsto (kx, k\sigma, ks)$ lies in Gau.

2. Markov-invariant metrics on finite simplices are c times the Fisher metric with $c > 0$ free [5].

3. The set $\{k_B, T, m\}$ contains no quantity of dimension length; a length requires an action or a pair of clock frequency and speed [4]. Thermodynamic length is invariant under a change of energy unit [6].

4. A flat metric is Euclidean in some chart; the chart need not be the physical one.

Proposition 12.2 (Counted channel removes dilation). *If F contains a counted channel whose law changes under a physical dilation, that dilation is not in Stab. If F contains only continuous channels whose relabeling groups include rescaling, dilations may lie in Stab.*

Proof. Lemma 4.4: no relabeling compensates a changed count law. $\square$

No-Go 12.3 (Scale-covariant count loss; [17]). If the anchor channel loses counts multiplicatively, $n' = (1 - \epsilon)n$ with ϵ unknown, the dilation fibre returns: Fisher rank 1 in (scale, ϵ). Additive slips and rate-dependent dead time do not reopen it. H-SI therefore includes a non-scale-covariant loss model.

12.2 Displacement and invariant metric

Definition 12.4 (Displacement). A displacement is an element of Γ acting on a target chart; concatenation is composition.

Imported Theorem 12.5 (Ratio scale; [9, 10]). *A monotone additive representation of concatenation is linear and unique up to one positive constant.*

Theorem 12.6 (Two-point invariant). *Suppose the latent space contains station and target positions in $\mathbb{R}^n$ and Gau contains the diagonal action of $SE(n)$, $n \geq 2$, or of $E(n)$, $n \geq 1$ (H- Γ). Then the orbit space of station-target pairs is $[0, \infty)$, parametrized by Euclidean separation ρ , and every natural two-point scalar is a function of ρ .*

Proof. The group is transitive on pairs at fixed separation: rotate (or, for $n = 1$, reflect) the difference vector, then translate. An invariant function is constant on orbits. $\square$

Theorem 12.7 (Metric proportionality). *Suppose relative Euclidean motions of the target lie in Γ , act transitively on the target chart, and have isotropy acting irreducibly on the tangent space $(H\Gamma)$. Then every natural metric module on the chart satisfies $\bar{g} = \lambda^2 h$, where h is the Euclidean metric of ρ and $\lambda > 0$ is constant.*

Proof. $\bar{g}$ and h are Γ -invariant (Theorem 5.3). The invariant symmetric forms of an irreducible compact isotropy form a line, since the commutant is $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$ and its symmetric part is $\mathbb{R}$; so $\bar{g} = \lambda(q)^2 h$. The ratio λ is a Γ -invariant function and Γ is transitive. $\square$

Theorem 12.8 (Anchor budget; [17]). *If the empirical isotropy is K , a constant metric on the target chart is fixed by exactly $\dim(\text{Sym}^2 V^*)^K$ radar anchors in general position: 1 under irreducible isotropy, $n(n+1)/2$ without symmetry.*

Proof. The admissible metrics form the linear space $(\text{Sym}^2 V^*)^K$, and each anchor is one linear equation. $\square$

Imported Theorem 12.9 (Unit-distance rigidity; [2]). *For $n \geq 2$, a map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ preserving distance one is an isometry. For $n = 1$ this fails: $x \mapsto x + 0.1 \sin 2\pi x$ preserves distance one and not distance one half.*

12.3 Anchor and identification

Hypothesis 12.10 (H-SI). The slice contains a two-way timing protocol that counts caesium cycles; c and $\Delta\nu_{\text{Cs}}$ take their exact SI values [47]; and round-trip counts add along concatenations for which ρ attains equality in the triangle inequality.

Proposition 12.11 (Radar distance). *Under H-SI the round-trip count n_{Cs} determines*

$$d_{\text{rad}} = \frac{c}{2\Delta\nu_{\text{Cs}}} n_{\text{Cs}},$$

and the factor $\frac{1}{2}$ is independent of the one-way synchronization convention [46]: with one-way speeds $c/(2\varepsilon)$ and $c/(2(1-\varepsilon))$ the round-trip time does not depend on ε .

Proposition 12.12 (Radar and rod distance). *On a static slice with metric representative $W^2(-dt^2 + dx^2)$ and a clock with $W = 1$ at the station, radar distance is independent of W away from the station, while rod distance is $\int W dx$. They agree on a path iff the exchanged-frequency ratio $W(x)/W(0)$ equals one along it. If translations along the path act in Γ on the clock channel, this ratio equals one.*

Proof. Null curves are conformally invariant, so the round-trip proper time at the station does not depend on W elsewhere. If a translation is in Stab , it maps the clock law at x to the clock law at $x + a$ with trivial relabeling on counts (Lemma 4.4), so the counted rates agree. $\square$

Hypothesis 12.13 (H-med). The propagation speed in the protocol is c , or is anchored by a protocol whose own anchor is exact, or the slowness field is resolved by a ray family with trivial null space.

Proposition 12.14 (Media). *Delay data are line integrals of slowness. Without H-med, thickness and speed are exchangeable (the pair (d, v) has a dilation fibre), and an incomplete ray family leaves a nonzero invisible slowness sector [14].*

Hypothesis 12.15 (H-stat). Time translation belongs to Γ on the source channel across an anchor outage.

Theorem 12.16 (Ergodic transfer; [17]). *Under $H\text{-stat}$, the second and third moments of a stationary source's invariant measure determine the detector map across the outage. The ambiguity is the symmetry group of those moments; if that group consists of isometries, all distances are determined. Drift is tested by a clock-based period invariant that no spatial map changes. Unflagged drift bounds the residual error by a measured floor.*

Hypothesis 12.17 ($H\text{-val}$). The proportionality constant λ fixed by one anchor predicts a second, held-out anchor within a declared tolerance.

Theorem 12.18 (Physical distance). *Assume $H\text{-reg}$, $H\text{-nuis}$, $H\text{-SI}$, $H\text{-med}$, $H\text{-val}$, the Γ -conditions of Theorems 12.6 and 12.7, and translation membership of Proposition 12.12. Then the unique Γ -invariant metric h has distance d_{rad} in metres, and for every natural experiment $\mathcal{E}$ on the chart*

$$\bar{g}_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h, \quad d_{\text{phys}} = d_{RG}^{\mathcal{E}} / \lambda_{\mathcal{E}},$$

with $\lambda_{\mathcal{E}}$ constant. RG angles equal physical angles. In rank one the identification additionally requires the concatenation law of Imported Theorem 12.5.

Proof. Theorems 12.6 and 12.7 give $\bar{g}_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h$ with h determined by ρ . Radar count is a natural two-point scalar, hence a function of ρ (Theorem 12.6); by $H\text{-SI}$ it is additive along collinear concatenations, hence linear (Imported Theorem 12.5). Proposition 12.11 fixes the constant in metres; Proposition 12.12 and $H\text{-med}$ identify radar with rod distance; $H\text{-val}$ excludes a response that fits one anchor only. In rank at least two Imported Theorem 12.9 makes one anchored distance sufficient. $\square$

13 The assembled theorem

Theorem 13.1 (Kernel–stabilizer theorem). *Let $\mathcal{S}$ be an observation system with joint law F , kernel groupoid $\mathcal{K}$ and effective symmetry Γ , and let all modules considered be natural. On each regular stratum:*

1. *exact ignorance is $\mathcal{K}$; infinitesimal ignorance is $V = \text{rad } g_F$;*
2. *every module descends to the quotient and is Γ -invariant;*
3. *under $H\text{-sound}$ and $H\text{-coh}$, completeness is constant on Γ_G -orbits;*
4. *clocks with equal kernels form one transition groupoid; a time flow in Stab makes it affine;*
5. *$SO^+(1, n) \leq \Gamma_q$, $n \geq 2$, forces the light cone and the conformal ray;*
6. *a counted channel excludes dilations from Γ ; Euclidean diagonal gauge and relative Euclidean symmetry force one two-point invariant and $\bar{g}_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h$;*
7. *under the hypotheses of Theorem 12.18, h is physical distance in metres.*

Corollary 13.2 (Compatibility equations). *On every regular stratum claimed for a slice the following hold simultaneously:*

$$\text{rad } g_F = V, \quad \text{isom}(\bar{g}_{\mathcal{E}}) = \text{isom}(h), \quad \Gamma_q C_q = C_q, \quad \lambda_{\mathcal{E}} \text{ constant},$$

the clock transition groupoid is Γ -invariant, and the held-out anchor is predicted. Each equation depends on the same subgroups of Γ ; if a subgroup is absent, every equation that uses it fails.

Corollary 13.3 (Stratified closure). *Under $H\text{-unif}$, the regular-stratum objects of Theorem 13.1 extend to the stratified quotient category of Definition 5.13:*

- *kernels are upper semicontinuous at walls;*

- *quotients specialize by maps under kernel inclusion, and by correspondences otherwise;*
- *symmetries descend through the normalizer of the wall kernel;*
- *distinguishability specializes under H -hell;*
- *characteristic sets are tame-typed, hyperbolically locked, or wild with window certificates;*
- *dynamical statements are typed certificates.*

Theorem 13.4 (Global transport). *1. The observable atlas closes without extra data. Transport is the transition groupoid within strata and specialization across walls: maps under kernel inclusion, correspondences otherwise. Wall geometry comes from re-discrimination of the specialized task family.*

2. Latent transport is undetermined without declared data (Theorem 10.7). Under H -cov it is canonical, coherent across walls, and curved exactly by the latent holonomy (Theorem 10.9).

No internal open problem remains. What remains is external (tame type enumeration) or an extension (Section 17).

14 The ladder

Dependencies point only to earlier rungs. Status: F forced, I imported, H hypothesis-conditioned (named), Γ Γ -conditional, N no-go, O open.

Rung	Statement	Status	Depends	Ref.
<i>Exact layer</i>				
L0	Observation system, joint law, relabeling group	input	—	2.1
L1	Kernel groupoid; $Q \cong \text{im } F$	F	L0	3.1
L2	Universal factorization; invariants are functions on Q	F	L1	3.2
L3	Joint kernel is the intersection; $Q \hookrightarrow \prod Q_i$ not onto	F	L1	3.3
L4	Garbling enlarges the kernel	F	L1	3.4
L5	Exact layer determines no higher structure	N	L2	3.5
<i>Symmetry layer</i>				
L6	$1 \rightarrow \text{Gau} \rightarrow \text{Stab} \rightarrow \Gamma \rightarrow 1$; Γ faithful on Q	F	L1	4.2
L7	The sequence need not split	N	L6	4.3
L8	Count rigidity	F	L0	4.4
L9	Orbit-quotient comparison	H: sound, compl	L1	4.6
L10	Rank, separation, completeness, reconstruction are distinct	N	L2,L9	4.7
L11	Coherent completeness	H: coh	L6,L9	4.8
<i>Differential layer</i>				
L12	Tangent sequence; infinitesimal completeness defect	H: reg	L2,L9	5.2
L13	Natural descent: basic, Γ -invariant, radical = V	F	L6,L12	5.3
L14	Descent of law-defined structures is automatic	F	L13	5.4
L15	Singular strata: slices and tangent cones	I	L12	5.5
L16	First-order tangent misses one-sided directions at walls	F	L15	5.6
L17	Kernel upper semicontinuity; pointwise convergence and sampling insufficient	H: unif	L2,L16	5.8
L18	Specialization map iff kernel inclusion; correspondence otherwise; functorial	F	L4,L17	5.10

Rung	Statement	Status	Depends	Ref.
L19	Symmetry descent through the normalizer of $\mathcal{K}_0$	F	L6,L18	5.11
L20	Tangent upper semicontinuity; rank lower semicontinuity; finite tame strata	F / I	L12,L17	5.12
L21	Stratified quotient category; holonomy within strata	H: unif	L18,L20	5.14
<i>Second-order layer</i>				
L22	Fisher is natural; $\text{rad } g_F = V$; Gaussian form	F	L13	6.2
L23	Regular reconstruction of (R, N, Σ)	I	L22	6.3
L24	Fisher does not determine the covariance	N	L23	6.4
L25	Second-order quotient has dimension $p + q + pq$	F	L2,L23	6.5
L26	Second-order shadow is not the law	N	L25	6.6
L27	Efficient metric: conventions and nuisance wall	H: nuis	L22	6.9
L28	Data processing contracts Fisher	I	L4,L22	6.10
L29	Kernel order is not Blackwell order	N	L4,L28	6.11
L30	Distinguishability specializes under Hellinger convergence; weak is not enough	H: unif, hell	L21,L28	6.13
L31	Quantum metrics: Petz family gate; commuting collapse	I	L22,L28	6.15
L32	Quantum task lock: SLD, BKM, WY	H: task	L31	6.17
L33	Nonparametric identification of the Petz function; bands, not panels	F	L31	6.18
L34	Quantum wall specialization iff $f(0) > 0$; re-discrimination	F	L30,L32	6.19
<i>Ignorance</i>				
L35	Persistent null; finite attainment	H: sens	L3,L22	7.2
L36	Sensor, precision and representation nulls	F	L35	7.3
L37	Finite-sample boundary reconstruction	I	L22,L27	7.4
L38	Rank walls are never certified from finite data	F	L20,L37	7.5
<i>Statistical distance</i>				
L39	RG distance: representative- and chart-free, Γ -invariant, dimensionless	H: reg, nuis	L13,L27	8.2
L40	Garbling does not increase RG length	F	L28,L39	8.3
L41	No angle on rank-one strata	N	L39	8.4
L42	Finite divergences are not additive	N	L39	8.5
L43	RG distance is experiment-relative	N	L39,L40	8.6
<i>Order and cone</i>				
L44	Fibre-compatible reachability descends	F	L2	9.1
L45	Projective typing of the cone	N	L44	9.3
L46	Realization branches	H: top	L15,L45	9.4
L47	Quadratic cone determines a conformal ray	I	L46	9.5
L48	Lorentz isotropy forces the light cone ($n \geq 2$)	Γ	L6,L13,L46	9.6
L49	Tame reduction: finite types, sheets, integer dimension	I / F	L20,L46	9.7
L50	No finite linear classification	N	L49	9.8
L51	Smoothness and strict convexity are dual	I	L46	9.9
L52	Hyperbolic lock; residual symbol implies ill-posedness	I / F	L49,L51	9.10
L53	Wild class: resolution invariants, window certificates only	N	L49	9.11
<i>Scale and transport</i>				
L54	Scale representative iff exact cocycle; holonomy	H: int	L47	10.2
L55	Strict transport fixes the conformal factor	F	L54	10.3
L56	Induced connection on regular strata	I	L12	10.4

Rung	Statement	Status	Depends	Ref.
L57	Entropy does not select a connection	N	L2	10.5
L58	Law-defined transport descends; latent transport iff projectable	F / H	L14	10.6
L59	No law-natural connection: vertical shears act transitively	N	L13,L58	10.7
L60	Canonical connection under declared covariance; curvature, memory, wall coherence	H: cov	L20,L59	10.9
<i>Duration</i>				
L61	Clock transition groupoid	F	L2	11.1
L62	Affine clocks under a stabilizer time flow	Γ	L6,L61	11.2
L63	Path independence as kernel inclusion; no orientation	F	L61	11.3
L64	Coherence duration and BCH memory	I	L61	11.4
L65	Finite-history noiseless experiments are stratified-regular	F	L21,L61	11.5
L66	Topological entropy from kernel counts	I	L2,L65	11.6
L67	Ruelle, Pesin and variational locks	I	L66	11.7
L68	Typed dynamical certificates; no zero-entropy certificate	F	L38,L67	11.8
L69	Cone fields of chaos: global only under uniform hyperbolicity	I	L52,L68	11.9
<i>Physical distance</i>				
L70	No internal unit	N	L22,L39	12.1
L71	Counted channel removes dilation	F	L3,L6,L8	12.2
L72	Multiplicative count loss reopens the dilation fibre	N	L71	12.3
L73	Ratio scale	I	L6	12.5
L74	Two-point invariant	Γ	L6,L13	12.6
L75	Metric proportionality $\bar{g} = \lambda^2 h$	Γ	L13,L74	12.7
L76	Anchor budget $\dim(\text{Sym}^2 V^*)^K$	Γ	L75	12.8
L77	Unit-distance rigidity ($n \geq 2$)	I	L75	12.9
L78	Radar distance; factor $\frac{1}{2}$	H: SI	L8,L61	12.11
L79	Radar equals rod distance iff no exchanged-frequency shift	Γ	L54,L71,L78	12.12
L80	Media: dilation fibre and invisible slowness	H: med	L35,L78	12.14
L81	Ergodic transfer of the metre across outages	H: stat	L62,L75,L80	12.16
L82	Physical distance	H: reg, nuis, SI, med, val; Γ	L70–L80	12.18
<i>Assembly</i>				
L83	Kernel–stabilizer theorem; compatibility equations	Γ	L1–L82	13.1
<i>Open</i>				
L84	Global transport: observable atlas closed; latent transport canonical under declared data	H: cov	L60,L83	13.4
L85	Enumeration of tame characteristic types (Hilbert’s sixteenth problem; external)	O	L49	17.2
L86	Non-static, quantum, non-dominated and infinite-dimensional systems	O	L83	17.3

15 Procedure

An application of RG to a physical slice consists of the following steps, each producing the object named.

1. Declare the protocols, the latent space and the relabeling group; form the joint law. (L0)
2. Determine the kernel by equality-of-law tests; obtain Q and its invariants. (L1–L3)
3. Test candidate generators of Stab by equality of laws after the declared operation and relabeling; record Γ . (L6)
4. Certify regular strata: constant rank of the law map, and of target and nuisance blocks; locate walls and use slices or jets there. (L12, L15, L27)
5. Compute natural modules: Fisher tensor, efficient metric, invariants, clock readouts, cone field. (L13, L22)
6. Read off the Γ -conditional structure: affine duration, cone type, two-point invariant, metric proportionality. (L48, L62, L74, L75)
7. Anchor: include a counted two-way timing channel; verify translation membership on the clock channel; resolve the medium. (L71, L78–L80)
8. At walls, specialize under declared uniform convergence (H-unif). Carry distinguishability only under Hellinger convergence. Report rank walls as upper bounds. (L17–L21, L30, L38)
9. For dynamics, use typed certificates on escalation plateaus, under the entropy locks. (L65–L69)
10. For quantum protocols, declare the task and identify the Petz function as a band; at support walls, re-discriminate. (L31–L34)
11. For latent transport, declare the second-order latent data or report the transport as undetermined. (L59, L60, L84)
12. Validate with a held-out anchor and check the compatibility equations. (L82, L83)

A failure at any step is attributed to one of: kernel misspecification; a wall; incomplete saturation; a smaller Γ ; a non-natural module; a failed anchor, medium or scale hypothesis; statistical rejection; or latent transport used without declared second-order data (L84).

16 Status

Layer	Status
Exact, symmetry, natural descent	forced
Strata and walls	forced under H -unif; distinguishability under H -hell; rank walls only bounded from data
Second order, Fisher, ignorance	forced or imported; nuisance under H -nuis
Quantum metric	family gate imported; member fixed by H -task; identified as a band from data
Cone and residual	realization branches; Lorentz under Γ ; residual tame-typed, hyperbolically locked, or wild (window certificates)
Scale, transport	scale under H -int; connection not law-natural; canonical under H -cov
Duration, dynamics	clock groupoid forced; dynamical statements as typed certificates
Physical distance	under H -reg, H -nuis, H -SI, H -med, H -val and Γ ; ergodic transfer under H -stat
Global	kernel-stabilizer theorem, stratified closure, global transport
Open	external enumeration of tame types; extensions (non-static, contextual quantum, non-dominated, infinite-dimensional)

17 Open problems

Remark 17.1 (Closed: latent connection). The former rung asking for a canonical latent connection is closed by Theorems 10.7 and 10.9. No connection is law-natural, and under declared second-order latent data the connection is canonical.

Open Problem 17.2 (Tame type enumeration). The residual branch of Theorem 9.4(5) is reduced by Section 9.1 to tame types (finite data), the hyperbolic lock, and the wild class (no finite classification). What remains is to enumerate the tame types of degree- d cones. For plane curves this is the first part of Hilbert’s sixteenth problem, known for $d \leq 7$ [41]; it is a problem of real algebraic geometry, not of RG.

Open Problem 17.3 (Extensions). Extend Theorems 12.18 and 13.1 to:

- non-static slices, where translations or rotations are absent from Γ and radar and rod distance differ;
- quantum protocols beyond the metric module, that is, joint laws of incompatible (contextual) measurement families;
- non-dominated or infinite-dimensional experiments.

Remark 17.4 (Computational witnesses). Each No-Go and each finite example in this paper has a deterministic numerical witness in the accompanying scripts, including the specialization, residual, chaotic, quantum and transport batteries [19, 18, 17, 20, 21]. The witnesses check the stated counterexamples; they are not premises of any proof.

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